

QUARTIC INVERSE-SPAN DYNAMICS: BINARY DEPTH JUMP AND EXACT FIBRES

ANONYMOUS

ABSTRACT. Let p be prime and send each $\mathbb{F}_p$ -subspace A of $\mathbb{F}_{p^4}$ to the span of the inverses of its nonzero points. Published work already classifies when patched inversion carries a subspace onto a subspace and describes inverse projective lines; those facts receive no contribution credit here. We use them to determine the resulting finite functional graph. Its sharp maximum tail is two at $p = 2$ and one at every odd prime: a non-subfield plane passes through a hyperplane only in the binary case. The recurrent states are zero, the full field, all lines, and all scalar copies of the quadratic subfield, and every period divides two. We give the exact depth enumerator, image stabilization, fixed-iterate counts, and dynamical zeta function. We also determine every target's fibre at every positive time. In particular, the 30 transient planes over $\mathbb{F}_2$ distribute uniformly over the 15 hyperplanes, two predecessors per target, before reaching the full field. A standalone exhaustive control checks 32,754 assertions for $p = 2, 3, 5$. This owner-dependent Round-0 note remains on external hold.

1. THE LITERAL MAP, PUBLISHED INPUTS, AND THEOREM CEILING

Fix a prime p , set $K = \mathbb{F}_{p^4}$, and let $\mathcal{X}_p$ be the lattice of all $\mathbb{F}_p$ -linear subspaces of K . For $A \in \mathcal{X}_p$, define

$$(1) \quad \mathcal{J}(A) = \text{span}_{\mathbb{F}_p}\{a^{-1} : a \in A \setminus \{0\}\}, \quad \mathcal{J}(0) = 0.$$

The span in (1) is part of the update. Powers $\mathcal{J}^t$ mean iteration on the subspace lattice, not multiplicative powers of its points.

Two published inputs set a hard ceiling on the claims. First, Kolomeec and Bykov classify the affine $\mathbb{F}_p$ -subspaces whose image under patched inversion $0 \mapsto 0$ is again affine: apart from the size-two boundary, they are exactly scalar copies of subfields [4]. Earlier inverse-closed and large-intersection results provide the algebraic context [2, 6]. Second, inverse projective lines in the cyclic field model are known to be normal rational curves in their spans; the small-field regime gives independent tuples [3, 5]. We use the classification as an external proposition and include a short denominator-clearing calculation for the line span. Neither input is a contribution of this note.

For $A \in \mathcal{X}_p$, let $\text{tail}(A)$ be the least time at which its orbit enters a cycle. Put

$$(2) \quad \begin{aligned} L &= p^3 + p^2 + p + 1, & P &= (p^2 + 1)(p^2 + p + 1), & Q &= p^2 + 1, \\ S &= 2 + 2L + P, & R &= 2 + L + Q, & F &= 2 + \gcd(2, L) + \gcd(2, Q). \end{aligned}$$

Thus L is the common number of lines and hyperplanes, P is the number of planes, and $S = |\mathcal{X}_p|$. For $t \geq 1$ and $B \in \mathcal{X}_p$, write

$$\nu_t(B) = |\{A \in \mathcal{X}_p : \mathcal{J}^t(A) = B\}|.$$

The finite-map zeta function is understood formally as

$$\zeta_{\mathcal{J}}(z) = \exp\left(\sum_{n \geq 1} |\text{Fix}(\mathcal{J}^n)| \frac{z^n}{n}\right),$$

following the periodic-point convention of Artin and Mazur [1].

Theorem 1 (complete quartic dynamics). *For the map (1), the following statements hold.*

(i) *The recurrent states are exactly*

$$(3) \quad 0, \quad K, \quad \xi \mathbb{F}_p, \quad \xi \mathbb{F}_{p^2} \quad (\xi \in K^\times).$$

The rank transition outside this set is

<i>input</i>	<i>number</i>	$\dim \mathcal{J}(A)$	<i>next state</i>
<i>non-subfield plane, $p = 2$</i>	$P - Q = 30$	3	<i>hyperplane</i>
<i>non-subfield plane, $p > 2$</i>	$P - Q$	4	K
<i>hyperplane</i>	L	4	K

On the recurrent scalar subfields, $\mathcal{J}(\xi \mathbb{F}_{p^d}) = \xi^{-1} \mathbb{F}_{p^d}$ for $d = 1, 2$; zero and K are fixed.
(ii) The maximum tail and depth enumerator $D_p(u) = \sum_{A \in \mathcal{X}_p} u^{\text{tail}(A)}$ are

$$(4) \quad \max_A \text{tail}(A) = \begin{cases} 2, & p = 2, \\ 1, & p > 2, \end{cases} \quad D_p(u) = \begin{cases} R + Lu + (P - Q)u^2, & p = 2, \\ R + (S - R)u, & p > 2. \end{cases}$$

Moreover,

$$(5) \quad |\mathcal{J}^t(\mathcal{X}_p)| = \begin{cases} S, & t = 0, \\ R + L, & p = 2, t = 1, \\ R, & p = 2, t \geq 2, \\ R, & p > 2, t \geq 1. \end{cases}$$

(iii) There are R recurrent states and F fixed states; the remaining $R - F$ recurrent states form $(R - F)/2$ two-cycles. Hence

$$(6) \quad |\text{Fix}(\mathcal{J}^n)| = \begin{cases} F, & n \text{ odd}, \\ R, & n \text{ even}, \end{cases} \quad \zeta_{\mathcal{J}}(z) = (1 - z)^{-F} (1 - z^2)^{-(R-F)/2}.$$

In particular, $F = 4$ for $p = 2$ and $F = 6$ for odd p .

(iv) Every positive-time target fibre is

$$(7) \quad \nu_t(B) = \begin{cases} 1, & B \in \text{Rec}(\mathcal{J}), B \neq K, \\ 2, & p = 2, t = 1, \dim B = 3, \\ 1 + L, & p = 2, t = 1, B = K, \\ 1 + L + P - Q, & p = 2, t \geq 2, B = K, \\ 1 + L + P - Q, & p > 2, B = K, \\ 0, & \text{otherwise.} \end{cases}$$

(v) Every transient state lies in the component of the fixed state K . For odd p , that component is a star whose leaves are all hyperplanes and non-subfield planes. For $p = 2$, its first level consists of the 15 hyperplanes, each with exactly two non-subfield-plane children. Every other fixed point or two-cycle is a bare component. Thus the number of weak components is $(R + F)/2$.

The residual theorem is the conjunction of the sharp characteristic dichotomy, the complete graph and image stabilization, and the all-time target fibres. The recurrent classification, inverse-line geometry, Gaussian counts, Singer action, and zeta conversion are treated as inputs or standard consequences. No novelty or priority inference is made.

2. RANK GROWTH AND THE INVERSE OF A PLANE

Lemma 2 (rank monotonicity). *For every $A \in \mathcal{X}_p$,*

$$(8) \quad \dim \mathcal{J}(A) \geq \dim A.$$

If equality holds, then

$$(9) \quad \mathcal{J}(A) = A^{-1} \cup \{0\}, \quad \mathcal{J}^2(A) = A.$$

Consequently, A is recurrent if and only if equality holds in (8), and every recurrent period divides two.

Proof. If $d = \dim A$, inversion gives $p^d - 1$ distinct nonzero elements inside $\mathcal{J}(A)$. An r -dimensional subspace has only $p^r - 1$ nonzero elements, so $r \geq d$. When $r = d$, cardinality forces the first identity in (9); patched inversion and a second span give the second.

Dimensions are nondecreasing along every orbit. They must be constant on a cycle, so recurrence forces equality at the first edge. Conversely, (9) places A on a cycle of length at most two. $\square$

We record the exact external input used with Lemma 2.

Proposition 3 (published inverse-image classification [4]). *Let E be a finite field of characteristic p , and patch inversion by $0^{-1} = 0$. If an affine $\mathbb{F}_p$ -subspace $A \subseteq E$ has $|A| > 2$, then its inverse image is an affine subspace if and only if $A = \xi \mathbb{F}_{p^d}$ for a subfield $\mathbb{F}_{p^d} \subseteq E$ and $\xi \in E^\times$.*

Proposition 3 receives zero contribution credit. It immediately rules out equality for a three-dimensional subspace of $\mathbb{F}_{p^4}$, since a subfield degree must divide four. The binary lines have size two, outside the stated hypothesis, but their behavior is direct.

The next calculation exposes exactly where the prime two differs. Its projective interpretation is already contained in inverse-line geometry [3, 5].

Lemma 4 (plane span). *Let A be a plane in K . Then*

$$(10) \quad \dim \mathcal{J}(A) = \begin{cases} 2, & A = \xi \mathbb{F}_{p^2}, \\ 3, & A \neq \xi \mathbb{F}_{p^2} \text{ and } p = 2, \\ 4, & A \neq \xi \mathbb{F}_{p^2} \text{ and } p > 2. \end{cases}$$

Proof. After scaling, write $A = \xi \langle 1, \alpha \rangle_{\mathbb{F}_p}$ and put $r = [\mathbb{F}_p(\alpha) : \mathbb{F}_p]$. Since $\alpha \in \mathbb{F}_{p^4} \setminus \mathbb{F}_p$, we have $r \in \{2, 4\}$. Projective representatives for the inverse points, after discarding the harmless scalar ξ^{-1} , are

$$(11) \quad 1, \quad (\alpha - t)^{-1} \quad (t \in \mathbb{F}_p).$$

Choose distinct $t_1, \dots, t_s \in \mathbb{F}_p$ with $s + 1 \leq r$. A relation

$$c_0 + \sum_{i=1}^s \frac{c_i}{\alpha - t_i} = 0$$

becomes $G(\alpha) = 0$ after multiplication by $D(\alpha) = \prod_i (\alpha - t_i)$, where

$$G(x) = c_0 D(x) + \sum_{i=1}^s c_i \frac{D(x)}{x - t_i}$$

has degree at most $s < r$. Minimality of r gives $G = 0$. Evaluating this polynomial identity at t_i yields $c_i \prod_{j \neq i} (t_i - t_j) = 0$, so every c_i and then c_0 vanish. Thus the span of (11) has dimension $\min\{p + 1, r\}$.

If $r = 2$, then $\langle 1, \alpha \rangle = \mathbb{F}_{p^2}$. If $r = 4$, the minimum is three at $p = 2$ and four for every odd prime. Restoring ξ proves (10). $\square$

Corollary 5 (rank transitions). *Every hyperplane maps to K . Every non-subfield plane maps to a hyperplane at $p = 2$ and to K at odd p .*

Proof. Lemma 2 leaves image dimension three or four for a hyperplane. Equality at three contradicts Proposition 3, so its image is K . Lemma 4 handles planes. $\square$

3. SHARP TIME AND THE RECURRENT CORE

Proof of Theorem 1(i)–(iii), apart from the image count at binary time one. By Lemma 2, a recurrent state of dimension at least two has a patched inverse image that is a subspace. Proposition 3 therefore makes it a scalar subfield. The subfield degrees inside four are 1, 2, 4. Lines, including binary lines, satisfy $\mathcal{J}(\xi \mathbb{F}_p) = \xi^{-1} \mathbb{F}_p$ directly; the same identity holds for scalar quadratic subfields. This proves (3). Corollary 5 supplies the remaining rows of the transition table.

There are L lines, P planes, and R hyperplanes. Scalar quadratic subfields are parametrized by $K^\times / \mathbb{F}_{p^2}^\times$, so their number is $(p^4 - 1)/(p^2 - 1) = Q$. This proves the state count S and recurrent count R in (2).

Every hyperplane has tail one. At an odd prime every non-subfield plane also has tail one, whereas at $p = 2$ it has tail two by Corollary 5. Since $P - Q = (p^2 + 1)(p^2 + p) > 0$, both bounds are attained. Counting these strata gives (4); it also shows that $\mathcal{J}^2(\mathcal{X}_2)$ and $\mathcal{J}(\mathcal{X}_p)$ for odd p lie in the recurrent core.

On recurrent lines, $\mathcal{J}$ induces inversion on the cyclic quotient $K^\times / \mathbb{F}_p^\times$, which has order L . On recurrent planes it induces inversion on the cyclic quotient $K^\times / \mathbb{F}_{p^2}^\times$, of order Q . Inversion on a cyclic group of order m has $\gcd(2, m)$ fixed points. Adding zero and K gives F fixed states; Lemma 2 allows no periods beyond two. Thus the other $R - F$ recurrent states form $(R - F)/2$ two-cycles. Since L and Q are odd at $p = 2$ and even at odd p , the displayed values of F follow.

The fixed-iterate count in (6) is now immediate. Taking the exponential of its defining series, or multiplying the standard factors for fixed points and two-cycles, gives the stated zeta function. $\square$

The time jump is therefore a projective-cardinality effect. It does not come from a different rule in characteristic two: the literal update (1) is uniform in p .

4. EVERY-TARGET FIBRES AND THE COMPONENT GRAPH

Only the binary plane-to-hyperplane row requires a target-resolved count.

Lemma 6 (twisted scalar symmetry). *For $\lambda \in K^\times$ and $A \in \mathcal{X}_p$,*

$$(12) \quad \mathcal{J}(\lambda A) = \lambda^{-1} \mathcal{J}(A).$$

The scalar action of $K^\times$ is transitive on the hyperplanes of K . Consequently, at $p = 2$ every hyperplane has exactly two non-subfield-plane predecessors.

Proof. Equation (12) follows by inverting every nonzero element before taking the span. The trace pairing on the separable extension $K/\mathbb{F}_p$ is nondegenerate. Hence every hyperplane is

$$H_c = \{x \in K : \text{Tr}_{K/\mathbb{F}_p}(cx) = 0\}$$

for some $c \in K^\times$, unique modulo $\mathbb{F}_p^\times$. Scalar multiplication sends these kernels transitively to one another.

At $p = 2$, Lemma 4 sends all $P - Q = 30$ non-subfield planes to hyperplanes. Twisted scalar symmetry gives bijections between the fibres over any two hyperplanes, so their sizes are constant. There are $L = 15$ targets, and $30/15 = 2$ sources lie over each. $\square$

Completion of Theorem 1(ii), (iv), and (v). Lemma 6 shows that every binary hyperplane occurs at time one. Together with the recurrent core this gives $|\mathcal{J}(\mathcal{X}_2)| = R + L$. Every such hyperplane maps to K , so the second image is exactly the recurrent core. At odd primes the transition table already sends every transient state to K , proving all of (5).

The restriction of $\mathcal{J}$ to the recurrent core is an involutive bijection. Every recurrent target other than K therefore has exactly one predecessor at every positive time. No transient state can be an additional predecessor: the transition table sends every transient orbit toward K .

A non-subfield plane has no predecessor. Indeed, a source of smaller rank cannot reach a plane, while a plane source with plane image would be an equality case and hence recurrent. At odd primes no update lands on a hyperplane. In the binary case Lemma 6 gives its two predecessors at time one; one further update sends every one of those paths to K , so hyperplane fibres vanish for $t \geq 2$.

At binary time one, the full field receives itself and all L hyperplanes, giving $1 + L$ sources. From time two onward it also receives all $P - Q$ non-subfield planes, giving $1 + L + P - Q$. At odd primes those planes arrive in one step, so the latter count holds for every $t \geq 1$. This proves (7).

The same argument identifies the components. Every transient state reaches K , and (7) shows that no transient tree attaches to any other recurrent cycle. For odd p all transient vertices are leaves. At $p = 2$, Lemma 6 gives the uniform two-level tree described in part (v). The recurrent core has F fixed cycles and $(R - F)/2$ two-cycles, so it has $(R + F)/2$ components. $\square$

5. EXACT CONTROLS AND SCOPE

The formulas specialize as follows. The depth column lists the numbers at successive tail depths, and the last column gives the full-field fibre at times one and two.

p	S	$ \text{im } \mathcal{J} $	R	F	cycles	depths	$\nu_{1,2}(K)$
2	67	37	22	4	$1^4 2^9$	22, 15, 30	16, 46
3	212	52	52	6	$1^6 2^{23}$	52, 160	161, 161
5	1120	184	184	6	$1^6 2^{89}$	184, 936	937, 937

A paper-local Python verifier uses only the standard library. It constructs an irreducible quartic for each of $p = 2, 3, 5$, implements field arithmetic, enumerates every subspace in reduced row-echelon form, and recomputes every directed edge. It then checks the recurrent scalar-subfield classification, rank transitions, cycles, depth histograms, images, twisted scalar symmetry, and every target fibre at times one through four. Two fresh processes match the frozen 32,754-assertion transcript byte for byte. These exhaustive finite checks are controls for implementation and boundary errors; the proofs above establish the formulas.

The scope is deliberately narrow. The base field is prime and the extension degree is four. The published inverse-image classification and inverse-line geometry remain zero-credit inputs, and the cycle/zeta count is a short consequence once the recurrent core is known. What remains in the note is only the exact temporal anomaly, complete graph and stabilization, and all-time target-fibre synthesis for the literal map (1). The bounded source search gives no novelty or priority conclusion. This anonymous Round-0 artifact is `GREEN_OWNER_THIN / HOLD_EXTERNAL`.

REFERENCES

- [1] Michael Artin and Barry Mazur. On periodic points. *Annals of Mathematics*, 81(1):82–99, 1965. doi: 10.2307/1970384.
- [2] Bence Csajbók. Linear subspaces of finite fields with large inverse-closed subsets. *Finite Fields and Their Applications*, 19(1):55–66, 2013. doi: 10.1016/j.ffa.2012.10.005.
- [3] Giorgio Faina, György Kiss, Stefano Marcugini, and Fernanda Pambianco. The cyclic model for $\text{PG}(n, q)$ and a construction of arcs. *European Journal of Combinatorics*, 23(1):31–35, 2002. doi: 10.1006/eujc.2001.0525.
- [4] Nikolay Kolomeec and Denis Bykov. On the image of an affine subspace under the inverse function within a finite field. *Designs, Codes and Cryptography*, 92(2):467–476, 2024. doi: 10.1007/s10623-023-01316-3.
- [5] Michel Lavrauw and Corrado Zanella. Geometry of the inversion in a finite field and partitions of $\text{PG}(2^k - 1, q)$ in normal rational curves. *Journal of Geometry*, 105(1):103–110, 2014. doi: 10.1007/s00022-013-0197-8.
- [6] Sandro Mattarei. Inverse-closed additive subgroups of fields. *Israel Journal of Mathematics*, 159(1):343–347, 2007. doi: 10.1007/s11856-007-0050-6.